



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	MODERN ORGANIC SYNTHESIS				
Type of Course	ELECTIVE IN MAJOR				
Semester	VIII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Preliminary idea about Reaction mechanism, reagents, organometallic chemistry, and Green methods				
Course Summary	This course explores various name reactions in organic chemistry, reagents, multistep synthesis, biochemical synthesis and modern trends in synthesis.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	To develop an understanding and application of different named organic reactions	U	C	Test /Seminar
CO2	To gain expertise in the usage and reactions of various reagents, focusing on catalysis and substitution	U	C	Discussion/ Assignment
CO3	To master the skills for multistep synthesis and biochemical synthesis with an emphasis on retrosynthetic analysis	An	C	Quizes/Test
CO4	To examine modern trends in synthesis, stressing on non-conventional methods,	Ap	E	Discussion/Seminar /Assignment

	catalysts and eco-friendly approaches			
CO5	To evaluate the role of organic compounds and reactions in pharmaceutical chemistry with open-ended, practical outcomes	Ap	P	Assignment/Test
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Named organic reactions		12	26
	1	Peterson Reaction, Julia Reaction, Ugi Reaction, Passerini reactions	1	
	2	Ene reaction, Ritter Reaction, Biginelli reaction, Skraup quinoline synthesis	2	
	3	Dess Martin oxidation, Baylis Hillman reaction, Eschenmoser–Tanabe fragmentation	2	
	4	Nazarov cyclization, McMurry coupling, Pauson–Khand reaction	2	
	5	Pummerer rearrangement, Ramberg–Bäcklund reaction, Rubottom oxidation, Rupe rearrangement	3	
	6	Staudinger reduction, Vilsmeier–Haack reaction, Wacker oxidation	2	
II	Reagents and reactions		12	26
	7	Reactions of carbene, umpolung, N-heterocyclic carbenes	1	
	8	Organocatalysis, Pd catalyzed reactions–Heck, Suzuki,	1	
	9	Suzuki, Stille Coupling	1	
	10	Cu catalyzed reactions- Buchwalds coupling	2	
	11	Synthesis of heterocycles by dipolar cycloaddition	2	
	12	Asymmetric hydrogenation	1	
	13	Alkenyl, allyl and aryl silanes and their substitution reaction	2	
	14	Silanes and organo boron reagents and their reactions.	2	
III	Multistep synthesis and biochemical synthesis		12	30

	15	Retrosynthetic analysis and Total synthesis of Longifoline (Corey, year),	3	
	16	Pencillin V (author, year) and cephalosporin (author, year), Tamiflu (author year)	3	
	17	Biosynthesis of mono and diterpenes,	2	
	18	Biosynthesis of morphine	2	
	19	Bio synthesis of lipids, fatty acids.	2	
IV	Modern Trends in Synthesis		12	16
	20	E factor, Non-conventional and eco-friendly reaction media- ionic liquids, supercritical CO ₂ .	4	
	21	Non-conventional energy sources- microwave, sonochemistry, electroorganic synthesis, visible light photocatalysis. Biomass valorisation, organocatalysis.	4	
	22	Reactions on solid acids bases, reactions on solid support, flow chemistry, enzyme catalyzed reactions, mechanochemistry, fluororous chemistry.	4	
V	Open ended-		12	
	23	Advanced spectroscopic methods/organo main group chemistry		

References

1. ORGANIC CHEMISTRY, by Jonathan Clayden, Nick Greeves, Stuart Warren, Oxford University Press.
2. March's Advanced Organic Chemistry: Reactions, Mechanisms and Structure, Michael B. Smith, Wiley.
3. Organic Chemistry Paperback, Francis Carey, McGraw-Hill Education
4. Advanced Organic Chemistry, David E. Lewis, Oxford Univ Pr; Illustrated edition
5. Principles of Organic Synthesis, R. O. C. Norman & J. M. Coxon, 3rd Ed., CRC Press, 2000.
6. Name reactions J J Li, Springer.
7. Spectrometric Identification of Organic Compounds, 8ed Paperback – 1 March 2022 Robert M. Silverstein, Francis X. Webster, David J. Kiemle, Wiley
8. Introduction to Spectroscopy, D. L. Pavia, 5th Ed., Cengage, 2015.
9. Organic Synthesis Special Techniques, V. K. Ahluwalia and Renu Aggarwal, Narosa Publishing house, Second Edn, 2015.
10. Organic chemistry, Stereochemistry and chemistry of natural products, I. L. Finar Vol. 2, Pearson.
11. Vogel's practical organic chemistry.

12. The Ritter Reaction: Trapping a Carbocation with a Nitrile, R. David Crouch, Journal of Chemical Education 1994 71 (8), A200, DOI: 10.1021/ed071pA200

13. Advanced methods of organic synthesis, W. Carruthers and Iain Colgham, Cambridge University Press; 4th edition (10 April 2015)

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	2		1	1	1	1	3		1		1		2
CO 2	2		1	1	1	1	3		2		3		3
CO 3	2	-	2	2	1	2	3		2		3		3
CO 4	2	-	2	2	2	2	3		2		3		3
CO 5	1	-	1	1	1	1	3		2		3		3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓